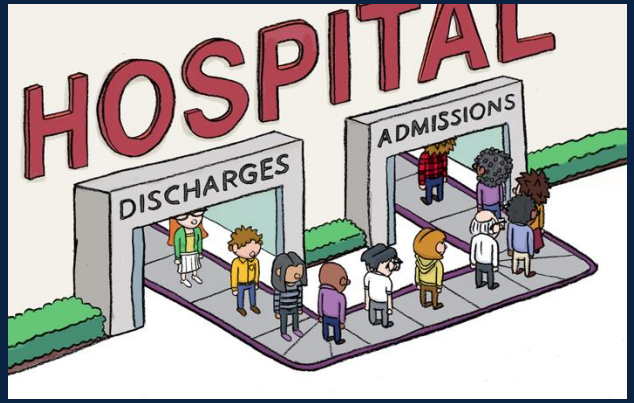
Diabetes Hospital Readmission Analysis



Identifying Early Readmission Risk in Diabetic Patients Using SQL + Machine Learning

Hugo Ortega-Morales · Databricks · SQL · Python · Scikit-learn · SHAP

PROBLEM STATEMENT

11%

of diabetic patients readmitted within 30 days

Early readmissions suggest treatment failure.
Hospitals face financial penalties under the **Hospital Readmissions Reduction Program (HRRP)** for excessive 30-day readmissions — making this both costly and a patient care issue.

Purpose: Build a risk flagging system that identifies high-risk diabetic profiles using clinical data available at the time of their hospital encounter.

Dataset: 101,766 encounters · 130 US hospitals · 1999–2008 · UCI ML Repository

APPROACH

01

SQL Exploration

CTEs, GROUP BY, CASE WHEN, HAVING across 9 analytical steps in Databricks

02

Statistical Testing

Chi-Square for categorical variables; Point Biserial Correlation for continuous features

03

Modeling

Logistic Regression → GridSearchCV hyperparameter tuning → Random Forest

04

Interpretability

SHAP feature importance to explain model predictions for clinical teams

05

Risk Flagging

Threshold-based system converting model scores into actionable patient flags

KEY FINDINGS

Prior Inpatient Visits — Dominant Predictor

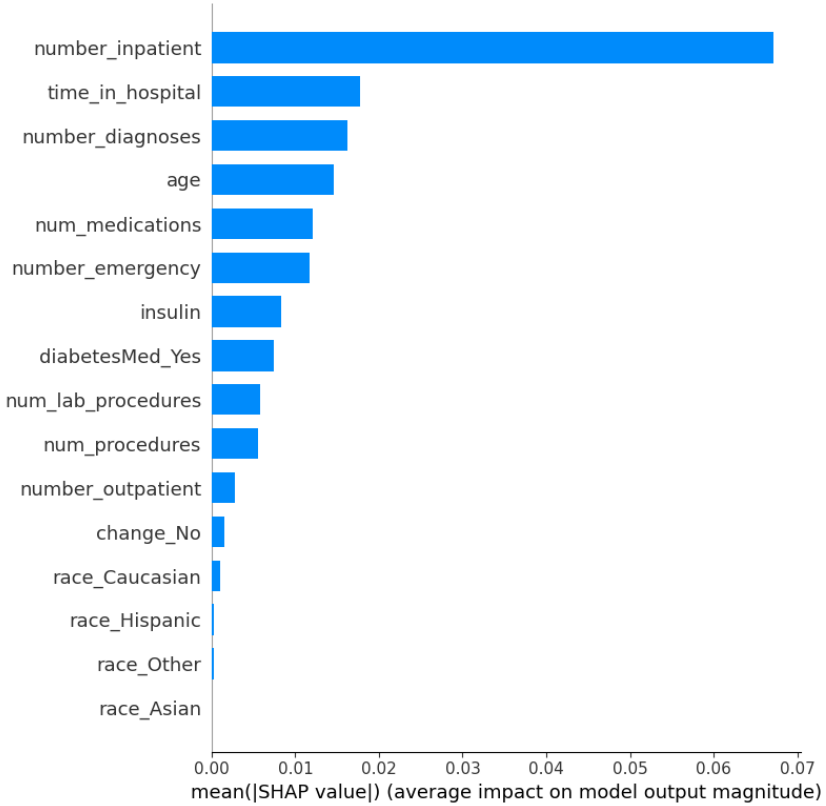
Readmission rate rises from 8% (0 prior visits) to 44% (8+ visits) — a 5x increase. SHAP confirms this as the single most influential feature, dwarfing all others.

Two Distinct High-Risk Profiles

Older patients (60–80) with multiple prior admissions on insulin show 20–28% rates. Young adults (20–30) show elevated rates (14.2%) unexplained by clinical features — likely complex factors outside the scope of the dataset.

Model Performance & Flagging System

Random Forest achieves 64% AUC. At 0.40 threshold the flagging system catches 85% of actual readmissions. Precision is limited — additional data is needed for clinical deployment.



RECOMMENDATION

1. Flag patients with 2+ prior inpatient visits for further discharge planning — regardless of model output. This is the strongest and most actionable finding.

2. Collect richer data: social determinants of health, medication compliance, and post-discharge data are needed to push model performance above 64% AUC.

3. Use the risk score system more as a tool, not a standalone decision. Scores above 0.40 should instigate clinician review, not automatic intervention.

GitHub: github.com/Hugoortega1025/diabetes-readmission-analysis